EXPERIMENT ASSESSMENT

ACADEMIC YEAR 2025-26

Course: Multidisciplinary Minor
Course code: MDC401
Year: SE SEM: IV

Experiment No. 10
AIM: Implement search functionality to filter API results dynamically.
Name: Prem Panchal
Roll Number: 48
Date of Performance:
Date of Submission:

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission.	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	5	3	2
Understanding	5	3	2
Journal work and timely submission.	10	8	4

Checked by

Name of Faculty: Dr. Yogesh Pingle

Aim: Implement search functionality to filter API search results dynamically.

Theory:

Modern web applications often handle large amounts of data retrieved from external APIs. Displaying all fetched data at once can reduce usability. To improve user experience, search and filtering functionality is implemented to dynamically narrow down results based on user input.

An API (Application Programming Interface) provides structured data in formats such as JSON. JavaScript fetches this data asynchronously and stores it in memory. A search input field allows users to enter keywords, which are used to filter the API results in real time.

1. Dynamic Search Filtering

Dynamic search filtering allows data to be filtered without refreshing the page. As the user types in the search box, JavaScript compares the input text with the fetched data and displays only matching records.

2. AJAX and Fetch API

The Fetch API is used to retrieve data asynchronously from the server. It returns a promise that resolves with the API response, which is then converted into JSON format. This ensures smooth and fast data retrieval.

3. JavaScript Array Methods

JavaScript provides built-in array methods such as filter() and includes() to efficiently search through data. These methods help compare user input with data fields like name, title, or description.

4. DOM Manipulation

The filtered data is displayed dynamically using DOM manipulation, ensuring instant updates to the user interface whenever the search input changes.

Algorithm / Procedure

1. Start the experiment.
2. Create an HTML file with a search input field.
3. Use JavaScript to fetch data from a public API.
4. Store the API response in an array.
5. Display all API results on the webpage.

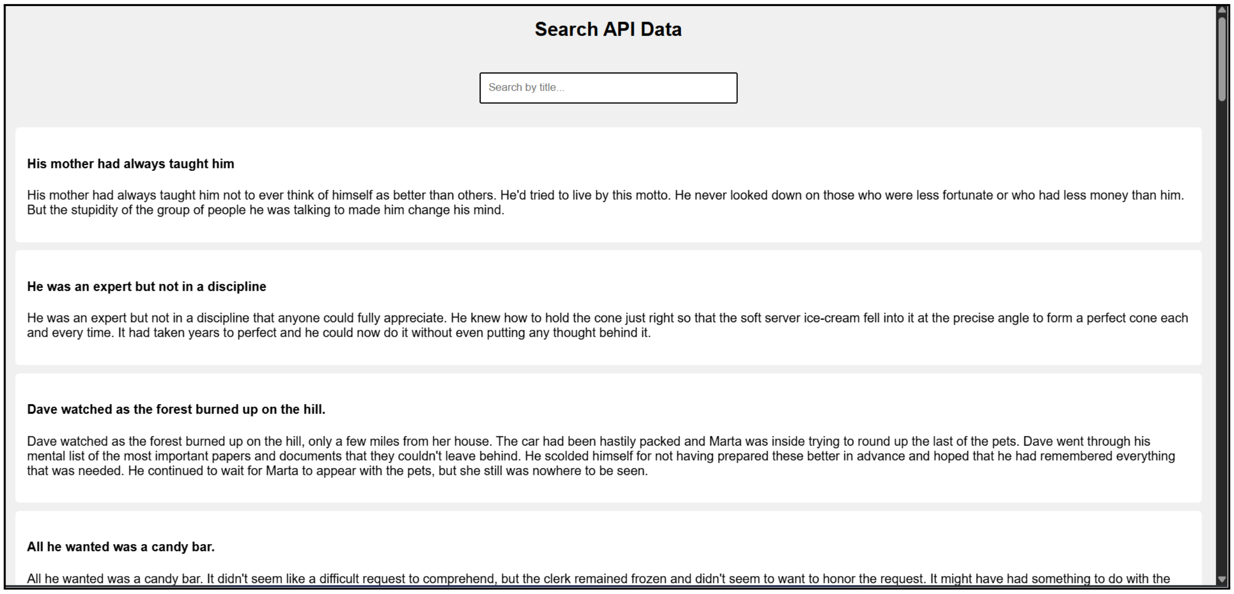
6. Capture user input from the search field.
7. Filter the stored data based on the search keyword.
8. Update the displayed results dynamically.
9. Repeat filtering as the user types.
10. Stop the experiment.

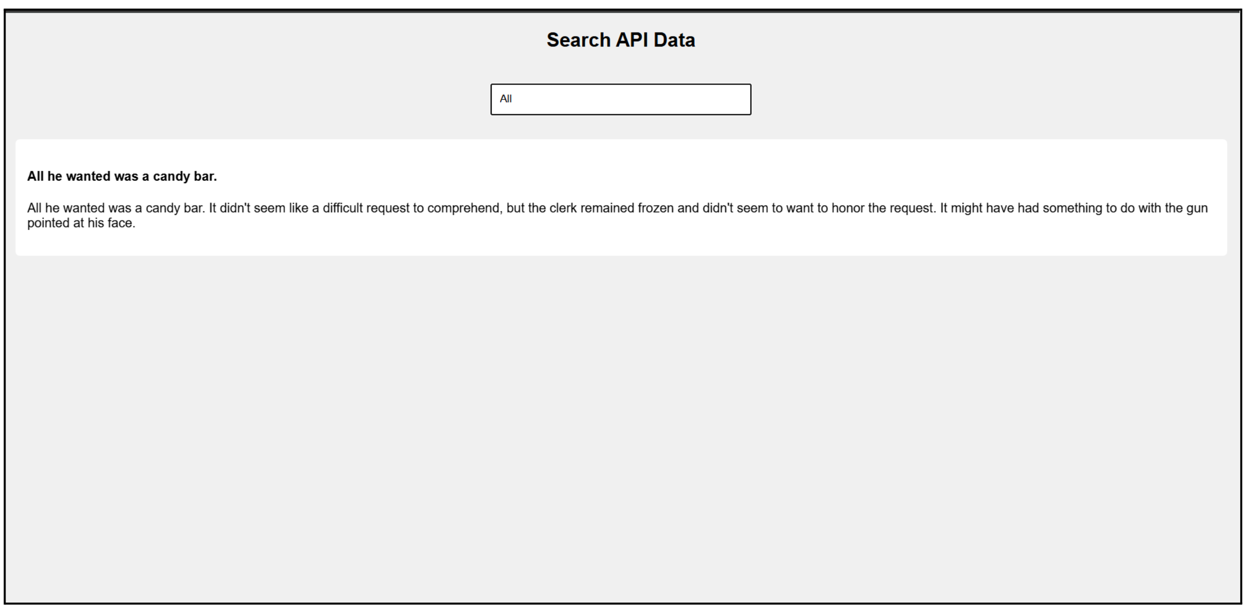
Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>API Search Filter</title>
  <style>
    body {
      font-family: Arial;
      background: #f4f4f4;
    }
    input {
      width: 300px;
      padding: 10px;
      margin: 20px;
    }
    .card {
      background: white;
      padding: 15px;
      margin: 10px;
      border-radius: 5px;
    }
  </style>
</head>
<body>

<h2 align="center">Search API Data</h2>
<center>
<input type="text" id="search" placeholder="Search by title...">
</center>

<div id="results"></div>

<script>
let apiData = [];
```

```
fetch("https://dummyjson.com/posts")
.then(response => response.json())
.then(data => {
  apiData = data.posts;
  displayData(apiData);
});

function displayData(data) {
  const resultDiv = document.getElementById("results");
  resultDiv.innerHTML = "";

  data.forEach(item => {
    resultDiv.innerHTML += `
      <div class="card">
        <h4>${item.title}</h4>
        <p>${item.body}</p>
      </div>
    `;
  });
}

document.getElementById("search").addEventListener("keyup", function() {
  const keyword = this.value.toLowerCase();
  const filteredData = apiData.filter(item =>
    item.title.toLowerCase().includes(keyword)
  );
  displayData(filteredData);
});
</script>

</body>
</html>
```

Output:


Search API Data
Search by title...
His mother had always taught him
His mother had always taught him not to ever think of himself as better than others. He'd tried to live by this motto. He never looked down on those who were less fortunate or who had less money than him. But the stupidity of the group of people he was talking to made him change his mind.
He was an expert but not in a discipline
He was an expert but not in a discipline that anyone could fully appreciate. He knew how to hold the cone just right so that the soft server ice-cream fell into it at the precise angle to form a perfect cone each and every time. It had taken years to perfect and he could now do it without even putting any thought behind it.
Dave watched as the forest burned up on the hill.
Dave watched as the forest burned up on the hill, only a few miles from her house. The car had been hastily packed and Marta was inside trying to round up the last of the pets. Dave went through his mental list of the most important papers and documents that they couldn't leave behind. He scolded himself for not having prepared these better in advance and hoped that he had remembered everything that was needed. He continued to wait for Marta to appear with the pets, but she still was nowhere to be seen.
All he wanted was a candy bar.
All he wanted was a candy bar. It didn't seem like a difficult request to comprehend, but the clerk remained frozen and didn't seem to want to honor the request. It might have had something to do with the


Search API Data
All
All he wanted was a candy bar.
All he wanted was a candy bar. It didn't seem like a difficult request to comprehend, but the clerk remained frozen and didn't seem to want to honor the request. It might have had something to do with the gun pointed at his face.